

Cadmium bioavailability from vegetable and animal-based foods assessed with in vitro digestion/caco-2 cell model.

BACKGROUND: Chronic dietary cadmium (Cd) exposure results in kidney dysfunction and decrease in bone mineral density. OBJECTIVE: To determine and compare the bioavailability of Cd from vegetable and animal-based foods. MATERIAL AND METHOD: Caco-2 cells were exposed to Cd in boiled pig kidney, ark shell, kale, raw kale, mixed boiled pig kidney with raw kale and CdCl₂ after in vitro digestion. Then cellular Cd uptake from the digests and reference CdCl₂ solution was measured by atomic absorption spectrometry. RESULTS: Cd bioavailability from animal-based foods was higher than that from vegetable-based foods. In addition, raw kale exhibited an inhibitory effect on Cd bioavailability when mixed with boiled pig kidney. However Cd in kale was increasingly absorbed after boiling. CONCLUSION: Cd binding to different molecular species, other food components in vegetable and animal-based foods, food combination, as well as cooking processes influenced the uptake of dietary Cd. A relative bioavailability factor accounted for the food matrix might be necessary for exposure assessment and consequently for estimation and prevention of the risk of dietary Cd.